KhanaKhazana: A Large-Scale Indian Food Dataset and Fine-Grained Classification Benchmark - Supplementary

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A Dataset Statistics

The corpus contains **1,760 classes** and **209,963 images** (mean 119.3 per class). The distribution is long-tailed, i.e. a short head (biryani, thali, tandoori-chicken, lassi, naan, dosa) exceeds 2,500 images each while most classes hold a few hundred or fewer (main paper Figure 2a). Native resolution, measured on a random 1,500-image sample, sits well above the 160×160 training resize for the great majority of images (main paper Figure 2b).

A.1 Visual Challenge Examples

The main paper's Table 3 defines four visual-challenge categories without illustrating them. Figure 1 shows corpus images for the two that carry measured support, i.e. each pair in panel (a) is annotated either with human top-1 (main paper Table 3) or with its model confusion from supplementary Table 3.

Dataset	Cuisine	Classes	Images
ChineseFoodNet [1]	Chinese	208	185,628
THFOOD-50 [2]	Thai	50	15,770
KenyanFood13 [3]	Kenyan	13	8,174
Sushi-50 [4]	Japanese	50	3,963
MyFood [5]	Brazilian	9	1,250
KhanaKhazana (ours)	Indian	**1,760**	**209,963**

Table 1: Cuisine-specific food-recognition datasets. All five prior datasets are single-cuisine benchmarks of a similar spirit to KhanaKhazana but at a much smaller vocabulary. The largest by class count (ChineseFoodNet, 208 classes) is still under an eighth of KhanaKhazana's label space, and none exceeds KhanaKhazana in image count.

A.2 Per-Tier Split Statistics

Table 2 gives split sizes and the *minimum* per-class image count for every Popular-N tier. The minimum falls from 48 images (100-200 classes) to 10 (300-600) and to a **single image** at 700-900 and 1,700 classes (two images at 1,000-1,500). The per-tier mean also varies substantially across the long tail. This accounts for the flat 37-41% region past ∼1,300 classes reported in the main paper, i.e. beyond 700 classes each tier admits classes with very few images (in the 1,700-class tier 197 classes have no training image at all), so tier-to-tier movement tracks which classes enter rather than class count itself.

Tier	Train	Test	Min imgs/class
Popular-100	12,813	2,750	48
Popular-200	19,468	4,185	48
Popular-300	26,267	5,644	10
Popular-400	33,093	7,110	10
Popular-500	38,890	8,353	10
Popular-600	42,886	9,215	10
Popular-700	69,701	14,964	**1**
Popular-800	77,510	16,637	**1**
Popular-900	84,713	18,168	**1**
Popular-1000	91,694	19,662	**2**
Popular-1300	110,373	23,678	**2**
Popular-1500	119,577	25,662	**2**
Popular-1700	119,581	25,862	**1**

Table 2: Split sizes for every Popular-N tier. The minimum per-class image count drops from 48 (100-200 classes) to 10 (300-600) and to a single image at 700-900 and 1,700 classes (two at 1,000-1,500), which is why accuracy past ∼1,300 classes flattens rather than continuing to fall.

B Full-Resolution Confusion Matrix

The main paper's Figure 4 shows this matrix at column width, where class labels are illegible by design. Figure 2 is the same matrix at full resolution with all 300 class names printed on both axes. Zoom in digitally to read them.

B.1 The 24 Strongest Off-Diagonal Confusions

Table 3 lists all 24 off-diagonal pairs at or above 35% row-normalized confusion; the main paper names five. Most are label-granularity artifacts, i.e. `oondees/kozhukkatta` and `pantua/ledikeni` are regional names for one dish, `milk-burfi/burfi` adds a redundant qualifier, `masala-chai/adrak-chai` differ by one spice, and `bhaji`, `cheesy-pav-bhaji` and `pav-bhaji` are three labels over one visual concept. Fewer reflect genuine ambiguity (`palak-sabji→aloo-methi`) or an underspecified base label (`korma`).

Every class here holds between 2 and 22 test images, so a single misrouted image moves the percentage by several points. These are evidence about *label design* in the long tail, not stable per-class error rates.



Figure 1: Real corpus images for the two visual-challenge categories with measured support. (a) Inter-class similarity: three pairs of distinct classes that are near-indistinguishable, each annotated with its measured human or model confusion. (b) Intra-class variability: six images of the single class *baingan bharta*, spanning the color range described in the main paper's Table 3.

True class	Predicted as	Conf.	n
oondees	kozhukkatta	100.0%	2
peda-sweet	milk-peda	83.3%	6
mango-rasam	rasam	71.4%	14
appam-veg-stew	kerala-veg-stew	66.7%	6
aloo-kulcha-on-tawa	kulcha	50.0%	8
palak-sabji	aloo-methi	45.0%	20
kati-roll	chicken-kathi-roll	44.4%	9
korma	chicken-korma	42.9%	7
korma	mutton-curry	42.9%	7
milk-burfi	burfi	42.9%	7
pantua	ledikeni	42.9%	14
aloo-paneer-paratha	aloo-paratha	41.7%	12
masala-chai	adrak-chai	40.9%	22
maacher-jhol	dal-chawal-chokka	40.0%	5
rabdi-khichdi	phirni	40.0%	5
zunka	makka-poha	40.0%	5
aloo-gobi-paratha	aloo-cheese-paratha	37.5%	8
aloo-gobi-paratha	gobi-paratha	37.5%	8
bhaji	pav-bhaji	37.5%	16
kala-chana-dosa	tips-dosa	37.5%	8
cheesy-pav-bhaji	pav-bhaji	36.4%	11
mutton-rogan-josh	mutton-curry	36.4%	11
pantua	gulab-jamun	35.7%	14
suji-ka-halwa	sheera	35.3%	17

Table 3: All 24 off-diagonal class pairs at or above 35% row-normalized confusion (DenseNet-121, Popular-300). "Conf." is the fraction of the true class's test images assigned to the predicted class, n is the true class's test-set size. `chicken-kathi-roll` abbreviates `chicken-kathi-roll-chicken-frankie`.

C Extended Comparison with Cuisine-Specific Food Datasets

Table 1 extends the main paper's Table 1 to five cuisine-specific datasets covering other geographies. Counts are as reported in each dataset's original publication.

D Extended Zero-Shot Model Comparison

Table 4 gives all eleven zero-shot models evaluated, ordered by top-1 accuracy; the main paper reports eight. SmolVLM-256M and SmolVLM-500M extend the range down by an order of magnitude and remain very low, with SmolVLM-256M essentially at chance. The ordering is not monotone in size, i.e. Qwen2.5-VL-3B falls below the older Qwen2-VL-7B and Llama-3.2-11B below both.

Evaluation protocol. All open-source models receive an identical prompt enumerating the 300-class vocabulary, i.e.

```
You are an expert in Indian cuisine. Look
at this food image and identify the dish.
The dish belongs to one of these 300
Indian food categories:
{comma-separated class list}
List your top 5 guesses in order, one
per line, numbered 1-5. Use the exact
category name from the list above. No
explanation.
```

Responses are parsed line-by-line and matched to the vocabulary by normalized string match, and unmatched guesses are discarded rather than counted correct. LLaVA-1.5-7B fails this step, returning bare list markers rather than dish names, which is why its 0.00% reflects instruction-following rather than absent domain knowledge.

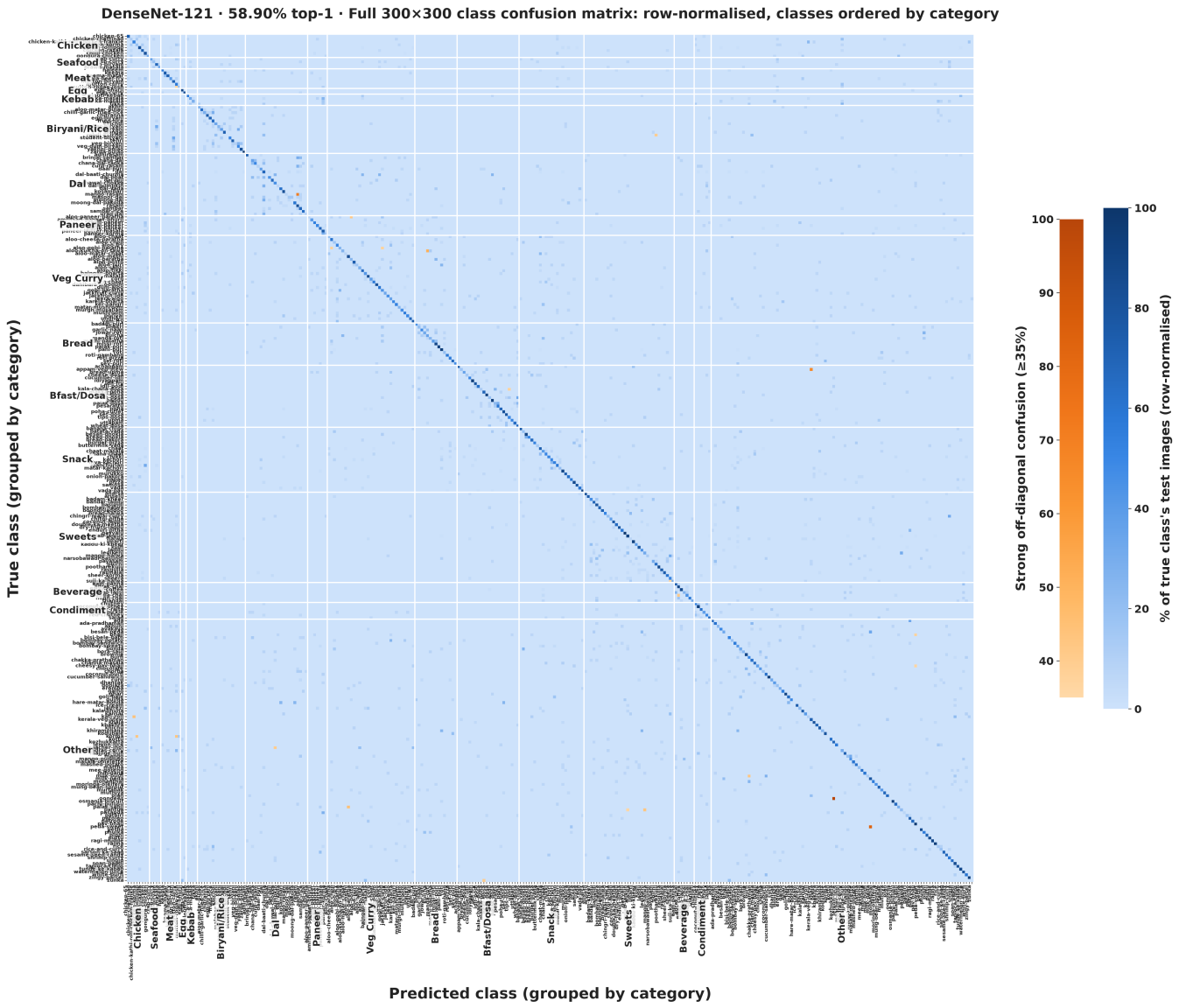


Figure 2: Full-resolution 300×300 confusion matrix (DenseNet-121, Popular-300 test split, 58.90% top-1), row-normalized, with all class names legible on both axes. Classes are grouped by 16 macro-categories, white lines mark category boundaries. Cells at or above 35% row-normalized confusion are highlighted in the warm ramp.

E Extended Coarse-Label Ablation: 88-Class Culinary Taxonomy

The main paper's coarse-label experiment bins dishes into 51 semantic groups and finds this *hurts* accuracy (49.16% at 51 classes, versus 75.45% at 100 fine-grained classes and 58.90% at 300). We repeated it at a second granularity to check the finding was not an artifact of that binning.

Setup and result. DenseNet-121 (ImageNet init) on an 88-class culinary taxonomy, i.e. 149,631 train / 18,702 validation images, 80 epochs, LR 1×10^{-4}, batch 64, cosine decay, label smoothing 0.1, and a `WeightedRandomSampler` against a 134× imbalance (smallest class 73 images, largest 9,808). Best validation accuracy was **54.74%** at epoch ~13, collapsing to 10.0% by epoch 80, i.e. the sampler oversampled the thinnest classes ~130×, consistent with the backbone memorizing the oversampled tail. The checkpoint, not the final epoch, is the reported result.

54.74% validation accuracy at 88 culinary classes (the 49.16% at 51 bins is a test score) sits above the 51-bin result and below the fine-grained results, reproducing the pattern at a second granularity with 3× more training data, i.e. *coarser, culinarily-defined labels do not raise accuracy, because culinary and visual taxonomy diverge.* The 88 bins are not visually coherent, e.g. one `mixed-veg-curry` bin spans 26 dissimilar dishes. A capped upsampling ratio, a minimum-image-count cutoff, or focal loss would avoid the collapse.

Model	Params	Top-1	Top-5
Random chance (1/300)	-	0.33%	1.67%
LLaVA-1.5-7B [6]	7B	0.00%	0.00%
CLIP ViT-B/32 [7]	151M	0.22%	2.28%
SmolVLM-256M [8]	256M	0.32%	0.51%
SmolVLM-500M [8]	500M	0.94%	1.04%
Qwen2-VL-2B [9]	2B	4.66%	7.69%
Llama-3.2-11B-Vision [11]	11B	8.97%	10.78%
Qwen2.5-VL-3B [10]	3B	9.67%	15.32%
InternVL2.5-4B [12]	4B	16.35%	22.52%
Qwen2-VL-7B [9]	7B	19.05%	26.42%
GPT-4o [13]	undisc.	39.95%	55.68%
Gemini 2.5 Flash Lite [14]	undisc.	48.25%	70.37%
Ours (DenseNet-121)	**8M**	**58.90%**	**84.85%**

Table 4: All eleven zero-shot models evaluated, ordered by top-1 accuracy. The three rows absent from the main paper's Table 4 are SmolVLM-256M, SmolVLM-500M and Qwen2.5-VL-3B. Parameter counts are the total published model size. Our fine-tuned DenseNet-121, at 8M parameters, outperforms every zero-shot model, including models hundreds to more than a thousand times larger.

F Training Recipe and Underfitting at Large Vocabularies

Table 5 lists the large-vocabulary tiers retrained on the final corpus with a single seed. The baseline recipe (80 epochs, learning rate 3×10^{-4}, label smoothing 0.1) underfits more and more as the vocabulary grows: final-epoch training accuracy (measured on augmented training batches) falls from 59.6% at 400 classes to 14.5% at 1,500, and validation accuracy exceeds it from 1,000 classes on. Test top-1 falls from 52.9% to 20.3% over the same tiers, which reflects the optimization budget as much as the vocabulary. The extended recipe (120 epochs, learning rate 10^{-4}, label smoothing 0.2) fits the large tiers far better: 62.5%, 49.5%, 43.8% and 42.8% training accuracy and 46.03%, 41.28%, 37.62% and 37.55% test top-1 at 1,000, 1,300, 1,500 and 1,700 classes. The main paper therefore uses the extended recipe for the retrained tiers of 1,000 classes and above. Results trained with different recipes are not comparable across rows.

G Limitations and Ongoing Work

Two limitations remain. First, no complete dataset-wide cropping or segmentation has been applied, i.e. some images are full photographs (plate, table, hands, garnish included) rather than isolated dish crops, only a pilot subset is cropped. Second, residual label noise remains across the long tail, and we do not measure inter-annotator agreement. Future work will develop a structured taxonomy of Indian foods at this scale to better capture regional, culinary, and compositional relationships. We also plan to explore custom food-domain embeddings to improve retrieval, representation learning, and fine-grained recognition across visually and semantically similar dishes.

H Cleaning and Verification Protocol

Table 6 lists every cleaning stage with the number of images it affected. The stages ran in the order below, with some overlap while new data was still being ingested.

Three to four annotators from our institute's annotation lab performed the manual review in a single pass, working directly in the interface. The "kept" count includes whole-class approvals, so it is not a count of individual inspections. We do not measure inter-annotator agreement, so the residual label error is unknown.

I Linear-Probe and Controlled-Experiment Details

I.1 Frozen-Encoder Linear Probes

We attach one linear layer to a frozen encoder and train only that layer, with the same recipe as the ViT linear probes of the main paper: 80 epochs, Adam, learning rate 3×10^{-4}, weight decay 10^{-3}, cosine schedule, label smoothing 0.1, batch size 16, bfloat16, the same augmentation, and the best epoch chosen by validation top-1. The test split is scored once afterwards. Table 7 lists the encoders. DINOv3 [15] features are the layer-normalized class token; Perception Encoder [16] features are the projected image embedding (1024-d). The probes are scored on 5,476 of the 5,644 Popular-300 test images, because some files were unavailable on the training machine. The reference fine-tuned DenseNet-121 (58.90%) was scored on 5,591 images.

I.2 Controlled Experiment and Seeds

Each controlled run trains DenseNet-121 (same recipe as the main Popular-300 run) on a random subset of n training images per class, drawn once with a fixed seed from the Popular-K training split. Validation and test splits are the tier's own. Training used seed 0. Table 8 gives all runs. The number of training images actually used is slightly below $K \times n$, and the test sets are slightly smaller than the tier manifests, because some files were unavailable on the training machine. The two extra seeds repeat the standard Popular-300 run (about 25.5K training images) with seeds 1 and 2.

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Classes	Recipe	Epochs / LR / smoothing	Train acc.	Val acc.	Test top-1	Test top-5	Test images
400	baseline	80 / 3×10^{-4} / 0.1	59.6%	52.4%	52.88%	78.96%	7,110
500	baseline	80 / 3×10^{-4} / 0.1	51.4%	47.7%	48.45%	76.33%	8,353
1000	baseline	80 / 3×10^{-4} / 0.1	30.4%	34.5%	36.29%	63.99%	19,662
1300	baseline	80 / 3×10^{-4} / 0.1	20.8%	27.0%	27.07%	54.20%	23,678
1500	baseline	80 / 3×10^{-4} / 0.1	14.5%	20.3%	20.27%	43.86%	25,662
1000	extended	120 / 10^{-4} / 0.2	62.5%	45.0%	46.03%	73.38%	19,662
1300	extended	120 / 10^{-4} / 0.2	49.5%	40.9%	41.28%	69.56%	23,678
1500	extended	120 / 10^{-4} / 0.2	43.8%	36.9%	37.62%	65.92%	25,662
1700	extended	120 / 10^{-4} / 0.2	42.8%	37.3%	37.55%	65.74%	25,862

Table 5: Final-epoch training and validation accuracy and test accuracy of the retrained large tiers (DenseNet-121, seed 0).

Stage	Method	Images affected
OCR label recovery	Text read from images filed under blog-author folders is matched against the class taxonomy and the image is moved to the matching dish class. In a later pass over an unidentified bucket we audited all 803 proposed existing-class destinations by hand, corrected 64, created 6 new classes, and moved the images.	12,540 recipe-blog images relocated; 2,628 moved out of the unidentified bucket (8,732 → 6,104)
Deduplication	Exact duplicates within a class and cross-class label conflicts are removed.	18,053 removed, of which 1,710 were cross-class conflicts
Non-food pruning	Manual review of cluster contact sheets from the Google-Maps candidate pool (restaurant interiors, people, menus), in two rounds. These images never entered the corpus.	110,873 pruned (109,358 + 1,515)
Outlier clustering	EfficientNet-B0 embeddings per class, two-cluster cosine k-means. A minority cluster is flagged when its centroid distance to the majority exceeds 0.05 and it holds at most 45% of the class. Classes with fewer than 10 images are not clustered.	46,483 images flagged in 1,216 classes
Manual review	A review interface shows each class as an image grid, flagged images first. The annotator keeps, deletes, crops, or moves each image to another class, or approves a whole class at once.	198,336 decisions in 1,681 classes: 168,062 kept (84.7%), 21,072 deleted (10.6%), 9,202 moved (4.6%)

Table 6: Cleaning stages and their effect. Review counts come from the decision log as of 23 August 2026.

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Encoder	Weights (Hugging Face)	Input	Norm.	Trainable / total	Top-1	Top-5
DINOv3-S/16	`facebook/dinov3-vits16-pretrain-lvd1689m`	224	ImageNet	0.12M / 21.7M	64.37%	88.02%
DINOv3-B/16	`facebook/dinov3-vitb16-pretrain-lvd1689m`	224	ImageNet	0.23M / 85.9M	70.93%	91.95%
PE-Core-B/16	`facebook/PE-Core-B16-224`	224	0.5 / 0.5	0.31M / 94.0M	68.86%	91.84%
PE-Core-L/14	`facebook/PE-Core-L14-336`	336	0.5 / 0.5	0.31M / 317.5M	75.66%	94.45%

Table 7: Frozen-encoder linear probes on Popular-300 (5,476 test images).

Run	Classes K	Images/class	Train images used	Test images	Top-1	Top-5
ctrl_k100_ipc5	100	5	485	2,661	40.40%	68.66%
ctrl_k100_ipc10	100	10	974	2,661	50.66%	78.39%
ctrl_k100_ipc20	100	20	1,945	2,661	56.71%	83.77%
ctrl_k100_ipc30	100	30	2,925	2,661	61.89%	86.58%
ctrl_k200_ipc5	200	5	985	4,137	29.39%	56.83%
ctrl_k200_ipc10	200	10	1,969	4,137	37.93%	65.29%
ctrl_k200_ipc20	200	20	3,940	4,137	49.07%	76.34%
ctrl_k200_ipc30	200	30	5,909	4,137	51.49%	78.73%
ctrl_k300_ipc5	300	5	1,463	5,476	24.85%	49.23%
ctrl_k600_ipc5	600	5	2,775	8,706	19.07%	40.79%
Popular-300, seed 1	300	all	25,524	5,476	57.23%	84.51%
Popular-300, seed 2	300	all	25,524	5,476	56.98%	84.50%

Table 8: Controlled class-count versus images-per-class runs and the two extra seeds of Popular-300 (DenseNet-121, one run each).

J Class-Level Human Benchmark

Table 9: Class-level human benchmark results (Popular-300 test split, $n \geq 2$ participants per class). "Acc." = human top-1; "IDK" = fraction of "I don't know" responses. Three patterns emerge, i.e. shape-cue classes achieved perfectly, visual-ambiguity failures (low IDK, low Acc., specific wrong label), and knowledge-gap failures (high IDK, regional dishes unfamiliar to most participants).

Pattern	Class	Acc.	IDK	Notes
Shape cue	*gulab-jamun*	100%	0%	Spheres in syrup
	pav-bhaji	100%	0%	Iconic street-food plating
	filter-coffee	100%	0%	Steel tumbler unmistakable
	rajma	100%	0%	Dark kidney beans
	momo	100%	0%	Dumpling shape universal
Visual ambiguity	*missi-roti*	0%	0%	Labeled *aloo-paratha*
	prawn-masala	0%	0%	Guessed shrimp/fish curry
	manchurian	33%	0%	Confused w. chilli-chicken
	paneer-korma	0%	0%	Collapsed into PBM
	phirni	0%	40%	Confused w. basundi/petha
Knowledge gap	*eromba*	0%	100%	Rare regional dish, unfamiliar
	galho	0%	25%	Rare regional stew, unfamiliar
	kerala-beef-fry	0%	40%	Regional, rare exposure

K Integration with the ITD Plate Pipeline

Table 10: Classifier integration results on ITD test segments (10,540 crops, 1,587 *thali* images). All DenseNet-121 rows use the same 1,700-class food-pretrained backbone (8M total parameters) as initialization, with its 1,700 corpus classes mapped onto the ITD label set (e.g. every curry to `non-spicy-curry`). The label set has 41 classes, a subset of the 50 ITD dishes. Trainable is the subset of those 8M parameters updated at each tier, not a separate model. † KNN uses a global prototype bank. ‡ Best at epoch 22.

Method	Trainable	Top-1
ITD KNN (reported) [17]	-	84.51%†
Zero-shot remap (173 slugs)	0	0.63%
Frozen backbone, 41-cls head	42 k	93.20%
Unfreeze denseblock4	2.2 M	97.86%
Unfreeze denseblock3	5.6 M	**99.17%**
Combined retrain (GT + web)	7.0 M	**99.50%‡**